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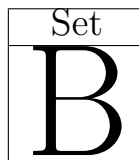
**Brac University**

Semester: Fall 2025

Course Code: CSE430

Digital Signal Processing

Faculty: EHQ



Assessment: *Quiz 4*
Duration: 25 minutes
Date: December 29, 2025
Full Marks : 20

$$e^{\pm j\omega t} = \cos(\omega t) \pm j \cdot \sin(\omega t)$$

■ Question 1 of 3 [8 marks]

Find the fundamental time period of the signals -

(a) [4 marks]

$$x[n] = 3.14 + 0.71 \cdot \sin\left(\frac{7\pi}{125}n + 33^\circ\right) + \cos\left(\frac{13\pi}{250}n - 16^\circ\right) - 0.24 \cdot \sin\left(\frac{7\pi}{125}n + 16^\circ\right)$$

(b) [4 marks]

$$x[n] = 0.43 + 0.45 \cdot \sin(0.125\pi n + 30^\circ) + 0.16 \cdot \cos(0.25n - 30^\circ)$$

■ Question 2 of 3 *[8 marks]*

Draw magnitude spectrum for the kernel $[1, 0, \underline{-2}, 0, 1]$.

The underlined value indicates the zero-th index. Maintain ranges for full marks.

■ Question 3 of 3 *[4 marks]*

Given you are applying Fourier analysis on the given time domain signal, what can you expect from the frequency spectrum? What would happen to the spectrum if the data beyond the range $[-2, 2]$ was made 0?

